

Advanced (Habanero)**Q1.** What is the expansion of $(a + b)^2$?

- (A) $a^2 + 2ab + b^2$
 (B) $a^2 - 2ab + b^2$
 (C) $a^2 + ab + b^2$
 (D) $a^2 - ab + b^2$
 (E) $a^2 + b^2$

Q2. Which of the following is a perfect square trinomial?

- (A) $x^2 + 4x + 4$
 (B) $x^2 - 4x - 4$
 (C) $x^2 + 2x - 4$
 (D) $x^2 - 2x + 4$
 (E) $x^2 + 4x - 4$

Q3. What is the factored form of $x^2 + 10x + 25$?

- (A) $(x + 5)^2$
 (B) $(x - 5)^2$
 (C) $(x + 5)(x - 5)$
 (D) $(x + 2)(x + 8)$
 (E) $(x - 2)(x - 8)$

Q4. What is the expansion of $(a - b)^2$?

- (A) $a^2 + 2ab + b^2$
 (B) $a^2 - 2ab + b^2$
 (C) $a^2 + ab + b^2$
 (D) $a^2 - ab + b^2$
 (E) $a^2 + b^2$

Q5. Which of the following is not a perfect square trinomial?

- (A) $x^2 + 6x + 9$
 (B) $x^2 - 6x + 9$
 (C) $x^2 + 2x - 4$
 (D) $x^2 - 2x + 4$
 (E) $x^2 + 6x - 9$

Q6. What is the factored form of $x^2 - 12x + 36$?

- (A) $(x + 6)^2$
 (B) $(x - 6)^2$
 (C) $(x + 6)(x - 6)$
 (D) $(x + 2)(x - 8)$
 (E) $(x - 2)(x - 8)$

Q7. What is the expansion of $(2a + b)^2$?

- (A) $4a^2 + 4ab + b^2$
 (B) $4a^2 - 4ab + b^2$
 (C) $4a^2 + 2ab + b^2$
 (D) $4a^2 - 2ab + b^2$
 (E) $4a^2 + b^2$

Q8. Which of the following is a perfect square trinomial?

- (A) $x^2 + 8x + 16$
 (B) $x^2 - 8x - 16$
 (C) $x^2 + 2x - 4$
 (D) $x^2 - 2x + 4$
 (E) $x^2 + 8x - 16$

Open Ended Questions (FRQ)**Q9.** Factor the expression $x^2 + 14x + 49$. Provide your work and final answer.**Q10.** Expand the expression $(3x - 2y)^2$. Provide your work and final answer.**Chef's Tips**

- Tip 1: Encourage students to use special patterns to simplify the expansions.
- Tip 2: Remind students to factor perfect square trinomials into the square of a binomial.
- Tip 3: Emphasize the importance of checking work by multiplying the factored form to ensure it matches the original expression.